

| Subsystem       | Device         | Function           | Wire Name | Assigned ESP32 Pin |  |
|-----------------|----------------|--------------------|-----------|--------------------|--|
| Power In        | Buck Converter | System Power       | 5V        | VIN (or 5V)        |  |
|                 | Buck Converter | Common Ground      | GND       | GND                |  |
| Attitude Sensor | MPU-6050       | Sensor Power       | VCC       | 3V3                |  |
|                 | MPU-6050       | Ground             | GND       | GND                |  |
|                 | MPU-6050       | I2C Data Line      | SDA       | GPIO 21            |  |
|                 | MPU-6050       | I2C Clock Line     | SCL       | GPIO 22            |  |
| Radio Receiver  | Receiver       | Receiver Power     | 5V        | Shared 5V Rail     |  |
|                 | Receiver       | Ground             | GND       | GND                |  |
|                 | Receiver       | UART Transmit      | TX        | GPIO 16 (RX2)      |  |
|                 | Receiver       | UART Receive       | RX        | GPIO 17 (TX2)      |  |
| Motor Outputs   | ESC            | Signal 1 (Rear-R)  | M1 PWM    | GPIO 13            |  |
|                 | ESC            | Signal 2 (Front-R) | M2 PWM    | GPIO 14            |  |
|                 | ESC            | Signal 3 (Rear-L)  | M3 PWM    | GPIO 27            |  |
|                 | ESC            | Signal 4 (Front-L) | M4 PWM    | GPIO 26            |  |
|                 | ESC            | Signal Ground      | Ground    | GND                |  |